

Elena Behzadi

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Education

Carnegie Mellon University, School of Computer Science

Expected May 2027

Bachelor of Science in Robotics

GPA: 3.64

Coursework: Control Systems, Computer Systems, Kinematics & Dynamics, SLAM, Path Planning, Autonomous Vehicles, Machine Learning, Computer Vision, Reinforcement Learning, Physics-Based Rendering

Experience

Applied Invention — Robotics Software Intern

May 2026 — August 2026

- **Built and deployed a custom 3D LiDAR SLAM pipeline on a production robot**; fused LiDAR/wheel odometry via a dynamics-informed Kalman filter, reaching **2 mm** relocalization over 50m
- Denoised time-of-flight camera depth data (phase-unwrapping error) via classical and deep-learning methods
- Built a color+depth segmentation and ICP pipeline reaching **0.5 mm** pick-and-place accuracy
- Implemented CUDA-accelerated sensor processing and real-time TCP data streaming

Kantor Lab, Carnegie Mellon University — Undergraduate Research Assistant

May 2025 — Present

- **Primary author**; paper peer reviewed accepted and presented at an international conference
- Designed fully autonomous leaf-detection and grasp planning for disease detection, reaching **95.8%** success
- Wrote custom multithreaded C++ servo-control software for synchronized **6-axis** gantry motion
- Built zero-shot leaf segmentation (SAM + RAFT-Stereo) reaching **97%** detection accuracy

Carnegie Mellon University — Teaching Assistant

Mar 2024 — Present

- *Robot Building Practices*: teach mechanical design and fabrication for **30+** students; led production of **400+** 3D-printed/CNC parts and design reviews. Led design and fabrication of robot chassis then built by students.
- *Concepts of Robotics*: developed problem sets for graph search and PID control

Projects

140 lb Autonomous Dodecacopter — *Large Air Vehicle Development*

Fall 2025

- **Led software team** for CMU's largest drone (**140 lb, 600 lb thrust**), designed for human transport
- Modeled vehicle dynamics in Isaac Sim; developed visual servoing and **sim-to-real** AirStack autonomy, tuning dual-GPS ArduPilot PID control
- Supported aircraft integration and flight testing, including wiring, ESC setup, and carbon-fiber assembly

F1TENTH Autonomous Racing

Spring 2026

- Implemented gap following, SLAM-based pure pursuit, and RRT* planning on Jetson of converted RC car.
- Identified dynamic bicycle model via nonlinear least squares; computed minimum-curvature raceline via convex QP, tuned online with Bayesian optimization

CMU Lunabotics — *Autonomy Lead, NASA Lunabotics Challenge*

2025 — Present

- Built path planning, SLAM, and navigation stack; ported LOAM-SLAM and Nav2 Hybrid A* to Jetson
- **Only rookie team in Lunabotics history to score autonomy points** at NASA's competition

Spectral Physically-Based Renderer

Spring 2026

- Built a C++ Monte Carlo renderer; added spectral path tracing with wavelength-dependent refraction and absorption for a stained-glass cathedral scene

Publications

E. Behzadi, A. Silwal, G. Kantor, *Vision-Guided Autonomous Leaf Sampling on T-Rex: Gantry Robot for Plant Disease Detection*, Earth Precision Agriculture Conference, Calgary, Canada, 2026

Technical Skills

Languages: C/C++, Python, MATLAB, Bash, Rust, JavaScript

Robotics: ROS/ROS2, MoveIt, Nav2, ArduPilot, Isaac Sim, SLAM, OpenCV, PCL, Gazebo

ML & GPU: PyTorch, TensorFlow, CUDA, RAFT-Stereo, YOLO, SAM

Tools: Linux, Git, Docker, Blender, CAD (SolidWorks/OnShape), PCB Design, Machining, CNC

Leadership & Activities

SDC Booth — Props Chair (physical game design, animatronics, fabrication)

Apex Buggy — team member, builds gravity-powered carbon fiber race vehicles